# Credit Risk Prediction Using Logistic Regression

**Lindokuhle Ncini, March 2026**

## Objective:

Predict whether a loan applicant is a **good or bad credit risk** using the German Credit dataset. Demonstrates full ML workflow: preprocessing, scaling, model training, evaluation, and visualization.

## Data Preprocessing:

- Numeric features (Age, Job, Credit amount, Duration) → **scaled** with StandardScaler
- Categorical features (Sex, Housing, Purpose, etc.) → **one-hot encoded**
- Missing values replaced with "unknown"
- Target variable Risk: good=0, bad=1

## Model & Evaluation:

- Logistic Regression (max_iter=1000) trained on **80% train / 20% test** split
- **Accuracy:** 96.5%
- **Confusion Matrix & Feature Importance:**

| **Predicted** | **Good (0)** | **Bad (1)** |
|---|---|---|
| Actual Good (0) | 130 | 0 |
| Actual Bad (1) | 7 | 63 |

## Figures:

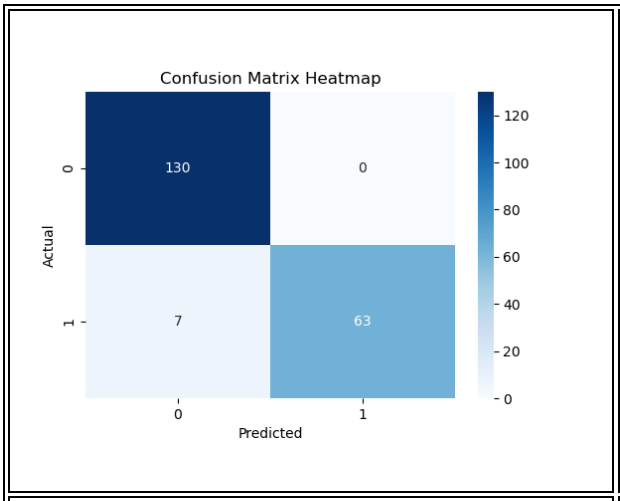


**Figure 1:** Confusion Matrix Heatmap

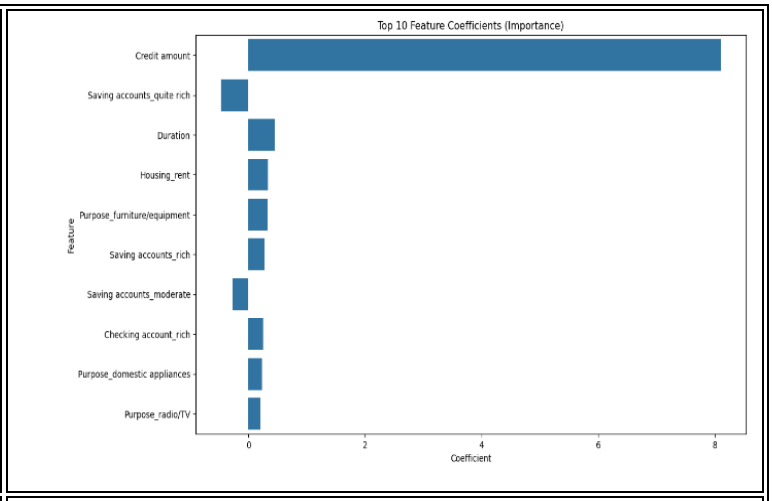


**Figure 2:** Top 10 Feature Coefficients

## Insights:

- **Credit amount** is the most important predictor; higher values → higher risk
- Numeric features dominate predictions; categorical features (Purpose, Housing) contribute moderately
- Model demonstrates strong accuracy and interpretability for decision-making